Practical 1 — Superstore product mix

Categorical tables, percentages and charts · 307312 Applied Statistics

- Complete Part 1 first.
- Companion to categorical summaries in Part 2.
- Allow two sessions of about 40–50 minutes.

By the end, you should be able to:

- Identify the observational unit and categorical variables.
- Build frequency, conditional-percentage and cumulative tables.
- Select charts with a clear business purpose.
- Explain why a change of denominator changes the answer.

1. The category manager's question

> Which product categories dominate activity, and does the mix differ across customer segments?

- **Business:** Superstore, a fictitious retailer of office products, furniture and technology.
- **Source:** Tableau's official sample workbook, downloaded 6 November 2026.
- **Period:** orders placed in 2023–2025.
- **Market:** United States only.
- **Unit:** one product line within an order; 6,682 lines across 3,322 orders.
- **Limit:** a teaching company; no claims about actual retailers or future demand.

Variable	Meaning	Treatment
`category`, `sub_category`	Product classification	Nominal
`segment`	Consumer, Corporate or Home Office	Nominal
`region`	Central, East, South or West	Nominal
`ship_mode`	Recorded service label	Nominal
`order_id`	Identifies an order	Identifier, repeated across product lines
`ship_days`	Days between order and dispatch	Ordered numerical duration, derived at order level

2. Import and check

- `data/business/superstore-lines.csv`; guide: `data/business/README.md`.
- Run the import cell by itself from either the project root or `extended-cases/superstore/`.

```
data_dir <- if (dir.exists("data")) "data" else if (dir.exists(file.path("..", "data")))
        file.path("..", "data") else file.path("..", "..", "data")
source(file.path(data_dir, "..", "lessons", "_includes", "course-tools.R"))
lines <- read.csv(file.path(data_dir, "business", "superstore-lines.csv"),
                  stringsAsFactors = FALSE, colClasses = c(postal_code = "character"))
orders <- make_orders(lines)
c(lines = nrow(lines), orders = nrow(orders), customers = n_distinct(lines$customer_id))
```

```
   lines    orders customers
    6682      3322       782
```

```
colSums(is.na(lines))
```

```
      row_id      order_id    order_date     ship_date     ship_mode
           0             0             0             0             0
 customer_id customer_name       segment       country          city
           0             0             0             0             0
       state   postal_code        region    product_id      category
           0             0             0             0             0
sub_category  product_name         sales      quantity      discount
           0             0             0             0             0
      profit return_status
           0             0
```

- `make_orders()` verifies consistent attributes and combines lines with the same order ID.
- It does not count a multi-line order several times.
- IDs and postal codes remain labels.

```
knitr::kable(head(lines[c("order_id", "category", "sub_category", "segment", "region")]))
```

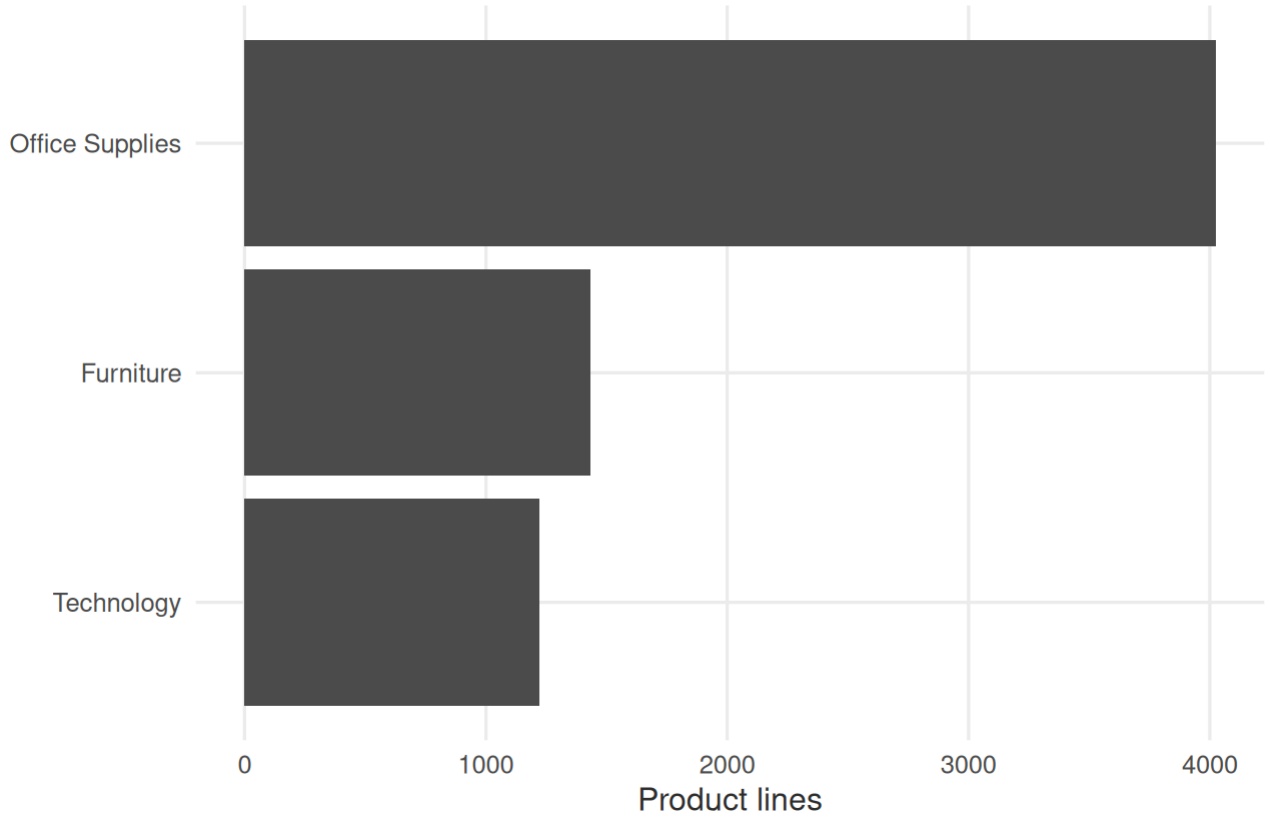
order_id	category	sub_category	segment	region
US-2023-103800	Office Supplies	Paper	Consumer	Central
US-2023-112326	Office Supplies	Binders	Home Office	Central
US-2023-112326	Office Supplies	Labels	Home Office	Central
US-2023-112326	Office Supplies	Storage	Home Office	Central
US-2023-141817	Office Supplies	Art	Consumer	East
US-2023-167199	Furniture	Chairs	Home Office	South

3. Describe the category mix

```
category_mix <- lines %>% count(category, name = "line_count") %>%
  mutate(line_pct = 100 * line_count / sum(line_count))
knitr::kable(category_mix, digits = 1)
```

category	line_count	line_pct
Furniture	1435	21.5
Office Supplies	4024	60.2
Technology	1223	18.3

```
ggplot(category_mix, aes(reorder(category, line_count), line_count)) +
  geom_col(fill = petra_ink) + coord_flip() +
  labs(x = NULL, y = "Product lines")
```



- `count()` produces frequencies.
- The denominator is all product lines in the extract.
- Office Supplies dominates line count; this does not establish the largest revenue or profit contribution.
- Categories have no required numerical order, so a cumulative category percentage would depend on an arbitrary arrangement.

Change the business measure

```
category_sales <- lines %>% group_by(category) %>%
  summarise(total_sales = sum(sales), total_units = sum(quantity), .groups = "drop") %>%
  mutate(sales_pct = 100 * total_sales / sum(total_sales))
knitr::kable(category_sales, digits = 1)
```

category	total_sales	total_units	sales_pct
Furniture	526612.5	5591	33.7
Office Supplies	472949.9	15230	30.2
Technology	564423.2	4576	36.1

- A line can contain several units and very different sales values.
- Line share, unit share and revenue share answer different questions.
- Technology leads revenue here even though it has fewer lines than Office Supplies.

Pause: Which measure would you use to discuss warehouse picking activity? Which would you use to discuss revenue? Explain the limits of each.

4. Compare customer segments

```
counts <- xtabs(~ segment + category, data = lines)
row_percent <- 100 * prop.table(counts, margin = 1)
knitr::kable(counts)
```

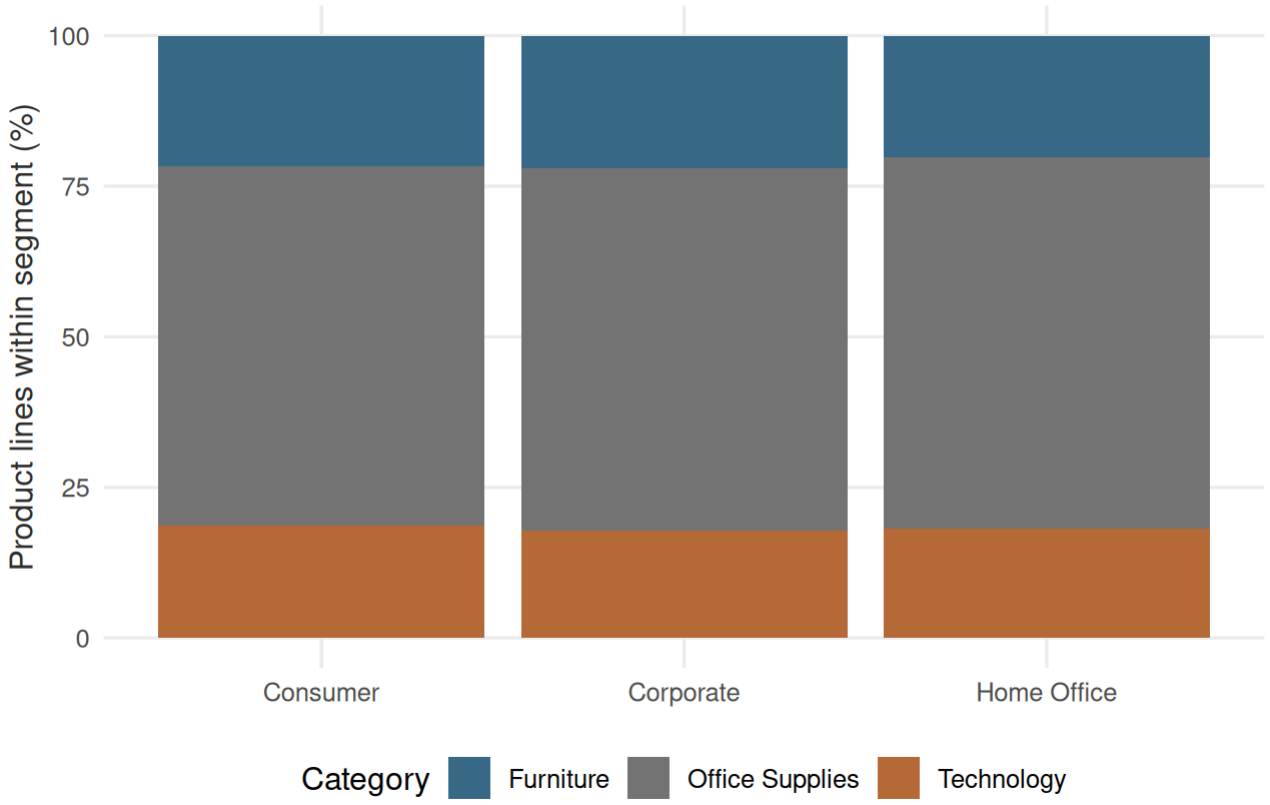
	Furniture	Office Supplies	Technology
Consumer	761	2106	656
Corporate	448	1229	363
Home Office	226	689	204

```
knitr::kable(row_percent, digits = 1)
```

	Furniture	Office Supplies	Technology
Consumer	21.6	59.8	18.6
Corporate	22.0	60.2	17.8
Home Office	20.2	61.6	18.2

- Rows: customer segments.
- Columns: product categories.
- Each percentage row totals 100%.
- A cell answers: “Within this segment’s product lines, what percentage belong to this category?”
- Dividing by a column total answers a different question: segment mix within a category.

```
plot_mix <- lines %>% count(segment, category) %>%
  group_by(segment) %>% mutate(percent = 100 * n / sum(n)) %>% ungroup()
ggplot(plot_mix, aes(segment, percent, fill = category)) +
  geom_col() + scale_fill_manual(values = petra_categories) +
  labs(x = NULL, y = "Product lines within segment (%)", fill = "Category")
```



- Every segment bar reaches 100%, even though segment counts differ.
- State both percentages and denominators when reporting a comparison.
- A difference of 5 percentage points is not automatically a 5% relative increase.

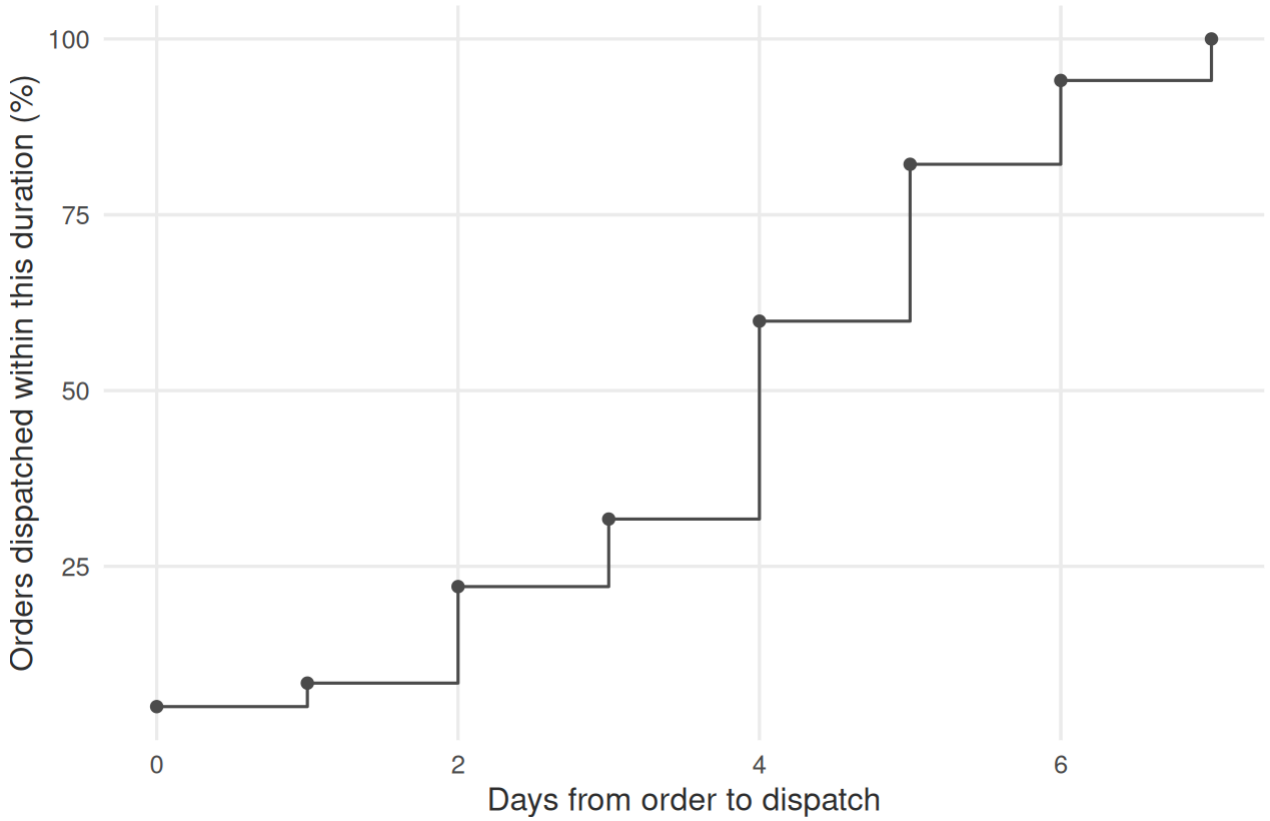
5. Use an ordered cumulative summary

> What share of orders is dispatched within each number of days?

```
shipping <- orders %>% count(ship_days, name = "order_count") %>%
  arrange(ship_days) %>%
  mutate(percent = 100 * order_count / sum(order_count), cumulative_pct = cumsum(percent))
knitr::kable(shipping, digits = 1)
```

ship_days	order_count	percent	cumulative_pct
0	168	5.1	5.1
1	111	3.3	8.4
2	456	13.7	22.1
3	319	9.6	31.7
4	935	28.1	59.9
5	741	22.3	82.2
6	396	11.9	94.1
7	196	5.9	100.0

```
ggplot(shipping, aes(ship_days, cumulative_pct)) +
  geom_step(colour = petra_ink) + geom_point(colour = petra_ink) +
  labs(x = "Days from order to dispatch", y = "Orders dispatched within this duration (%)")
```



- Durations have a natural order, so cumulative percentages are meaningful.
- Use one record per order; counting lines would give large baskets extra weight.
- Dispatch time is not delivery time.
- Observed lag is not the same as an advertised service guarantee.

6. Interpret a returns table correctly

```
return_mix <- orders %>% count(return_status, name = "order_count") %>%
  mutate(percent = 100 * order_count / sum(order_count))
knitr::kable(return_mix, digits = 2)
```

return_status	order_count	percent
Listed return	191	5.75
Not listed	3131	94.25

- An order is **Listed return** when its ID appears in the workbook's Returns sheet.
- The unit and denominator are orders, not products, returned units or customers.
- **Not listed** is absence from this source table; it is not independently confirmed non-return.
- Do not use the table to infer refund amounts or to adjust recorded profit.

7. Independent business briefing

The regional manager asks for a one-page summary.

1. Build a region frequency table using **orders**, with percentages.
2. Compare shipping-mode percentages **within each region**.
3. Show one chart that makes this comparison readable.
4. Calculate the percentage dispatched within four days, overall and by region.
5. Report the listed-return percentage for each segment, using orders.
6. Write four bullets: finding, numerical evidence, denominator and limitation.

```
# Use orders for these questions.
# Start with count(region), then xtabs(~ region + ship_mode, data = orders).
# For a group proportion: mean(ship_days <= 4) within each region.
```

Check your reasoning:

- Would line-based results give every order equal weight?
- Are you interpreting dispatch rather than delivery?
- Does the reader know whether a percentage is within a row or within the full table?

8. Continue

- Keep your tables and briefing in your RStudio project.
- Continue to Practical 2: Order size.
- Source definitions and extract limits: `data/business/README.md`.